Continuous-Time Portfolio Optimization: The Merton Problem and the Cost of Estimation Risk

Group project plan and work division

MAC F244 Stochastic Calculus and Applications in Finance, Semester I, 2026–27

Proposal due **31 Oct 2026** · Final submission **15 Nov 2026**

#	Role	Name	ID
1	Theory and coordination	Rishi Kumar Motwani	2024ADPS0838G
2	Data and parameter estimation	_____	_____
3	Simulator and verification	_____	_____
4	Estimation-risk experiments	_____	_____
5	Real-data backtest	_____	_____

How to use this document

- **Everyone** reads Sections 1–4 (the project, the maths, the fixed settings, the shared code). Then read your own section.
- If you give this PDF to an AI assistant to help with code: give it the *whole* PDF, tell it which person you are, and ask it to follow Section 4 *exactly* (file names, function names, arguments). Otherwise our pieces will not fit together.
- **You must understand what your code does.** The viva (15–20 min) is about the mathematical formulation and key findings, and all five of us sit it together. Every member must be able to explain the derivation of π^* (Section 2) and their own results.

1 The project in one page

Problem statement. An investor splits wealth between a risky stock and a risk-free bank account and can rebalance continuously. *What fraction of wealth should be in the stock to maximise expected utility of final wealth?* This is the Merton problem (topic 14 in the course list). Its answer is a simple formula, $\pi^* = (\mu - r)/(\gamma\sigma^2)$, but the formula needs the *true* drift μ and volatility σ, which nobody knows. So our real question is:

Our research question

Does Merton's optimal strategy still work when its parameters must be estimated from real market data, and how much does the estimation error cost?

The story of the project (one connected flow).

1. **Theory** (Person 1). Write the model, derive the HJB equation, solve it, and get π^*. Derive a formula for the cost of using a wrong fraction.
2. **Data** (Person 2). Estimate μ, σ, r from NIFTY 50 and three NSE stocks. Show that σ is estimated precisely but μ is not.
3. **Simulator** (Person 3). Build an Euler–Maruyama simulator, check it is accurate, and use it to confirm numerically that π^* really gives the highest expected utility *when μ, σ are known.*
4. **Estimation risk** (Person 4). In a pretend market where we *choose* the true μ, σ (so the true π^* is known), generate 1,000 fake price histories, re-estimate $\hat{\pi}$ from each, and measure how wrong $\hat{\pi}$ is and how much utility that loses.

5. **Backtest** (Person 5). On real prices, each month estimate $\hat{\pi}$ from past data only, invest with it for one month, roll forward, and compare with simple rules ($\pi = 0,\ 0.6,\ 1$).

Expected outcome. Inside the model, π^* is optimal. In practice, μ is so noisy that $\hat{\pi}$ can range from negative to well above 1, and the average utility lost may exceed the entire benefit of investing optimally; the estimated Merton strategy may then do no better than (or worse than) a fixed 60/40 rule. We quantify this precisely. This satisfies the course requirements of real data, effects of changes in data and assets, and analysis beyond reproducing formulas.

2 The mathematical model (shared notation)

Everyone uses exactly this notation. Time is measured in **years**; all rates are **annual, continuously compounded, written as decimals** ($6\% = 0.06$).

Symbol	Meaning
S_t	stock (or index) price at time t
B_t	bank account value; grows at the risk-free rate r
$\mu,\ \sigma$	drift (expected return per year) and volatility of the stock
W_t	standard Brownian motion (the source of randomness)
π_t	fraction of wealth held in the stock; $1-\pi_t$ is in the bank. $\pi > 1$ means borrowing, $\pi < 0$ means short selling
X_t	investor's wealth
$\gamma > 0$	relative risk aversion (larger γ = more cautious). Main value $\gamma = 3$
T	investment horizon; L = length of the data sample in years
Δt	time step; one trading day is $\Delta t = 1/252$

2.1 Market and wealth

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t \quad \text{(geometric Brownian motion)}, \qquad dB_t = rB_t\,dt. \tag{1}$$

By Itô's lemma, $d\ln S_t = (\mu - \tfrac{1}{2}\sigma^2)\,dt + \sigma\,dW_t$, so $S_t = S_0 \exp\big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big)$.

If the investor holds ϕ_t shares and ψ_t units of the bank account, $X_t = \phi_t S_t + \psi_t B_t$. *Self-financing* (no money added or removed) means $dX_t = \phi_t\,dS_t + \psi_t\,dB_t$. Writing $\pi_t = \phi_t S_t / X_t$ gives the **wealth SDE**

$$dX_t = X_t\big[r + \pi_t(\mu - r)\big]dt + X_t\,\pi_t\,\sigma\,dW_t. \tag{2}$$

2.2 Objective

CRRA (power) utility $U(x) = \dfrac{x^{1-\gamma}}{1-\gamma}$ ($\gamma \neq 1$). The investor solves

$$V(t,x) = \sup_{\pi}\ \mathbb{E}\big[\,U(X_T)\ \big|\ X_t = x\,\big].$$

2.3 HJB equation and its solution

Dynamic programming plus Itô's lemma on $V(t, X_t)$ give the Hamilton–Jacobi–Bellman equation

$$0 = V_t + \sup_{\pi}\Big\{\big[r + \pi(\mu - r)\big]xV_x + \tfrac{1}{2}\pi^2\sigma^2x^2V_{xx}\Big\}, \qquad V(T,x) = U(x). \tag{3}$$

The bracket is a downward parabola in π (since $V_{xx} < 0$). Setting its derivative to zero:

$$\pi^* = -\frac{(\mu - r)\, V_x}{\sigma^2\, x\, V_{xx}}.$$

Guess $V(t,x) = f(t)\,\dfrac{x^{1-\gamma}}{1-\gamma}$. Then $V_x = f x^{-\gamma}$, $V_{xx} = -\gamma f x^{-\gamma-1}$, so $xV_{xx}/V_x = -\gamma$ and

Merton's answer

$$\pi^* = \frac{\mu - r}{\gamma\,\sigma^2} \qquad \text{(constant: independent of time and wealth)}$$

Substituting back: $f(t) = \exp\Big((1-\gamma)\big(r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\big)(T-t)\Big)$.

2.4 Certainty equivalent and the cost of a wrong π

For a *constant* π, (2) is a GBM, so $X_T = x_0 \exp\big((r+\pi(\mu-r)-\frac{1}{2}\pi^2\sigma^2)T + \pi\sigma W_T\big)$ with $W_T \sim N(0,T)$. Computing $\mathbb{E}[X_T^{1-\gamma}]$ (moment of a lognormal) gives $\mathbb{E}[U(X_T)] = U\big(x_0 e^{\mathrm{CE}(\pi)\,T}\big)$ where

$$\mathrm{CE}(\pi) = r + \pi(\mu - r) - \tfrac{1}{2}\gamma\sigma^2\pi^2. \tag{4}$$

Meaning: CE is the guaranteed yearly return that the investor values exactly as much as the risky strategy. It is a parabola in π with peak at π^*, so

Loss from using $\hat{\pi}$ instead of π^* (per year)

$$\mathrm{Loss}(\hat{\pi}) = \mathrm{CE}(\pi^*) - \mathrm{CE}(\hat{\pi}) = \tfrac{1}{2}\gamma\sigma^2\,(\hat{\pi} - \pi^*)^2, \qquad \underbrace{\mathrm{CE}(\pi^*) - r}_{\text{benefit of optimal investing}} = \frac{(\mu-r)^2}{2\gamma\sigma^2}.$$

Worse than cash: $\mathrm{CE}(\hat{\pi}) < r$ exactly when $\hat{\pi} < 0$ or $\hat{\pi} > 2\pi^*$.

2.5 Estimating μ and σ from data

Daily log returns $R_i = \ln(S_i/S_{i-1})$ are i.i.d. $N\big((\mu - \frac{1}{2}\sigma^2)\Delta t,\ \sigma^2\Delta t\big)$ under GBM. With n returns, sample mean m and sample variance s^2:

$$\hat{\sigma}^2 = \frac{s^2}{\Delta t}, \qquad \hat{\mu} = \frac{m}{\Delta t} + \frac{\hat{\sigma}^2}{2}, \qquad \hat{\pi} = \frac{\hat{\mu} - r}{\gamma\hat{\sigma}^2}. \tag{5}$$

Precision ($L = n\Delta t$ years of data): $\mathrm{se}(\hat{\sigma}) \approx \sigma/\sqrt{2n}$ (small, improves with more observations), but $\mathrm{se}(\hat{\mu}) \approx \sigma/\sqrt{L}$, which depends only on the *number of years*, not on how often we sample. Hence

Theoretical predictions that Person 4 will test

$$\mathrm{sd}(\hat{\pi}) \approx \frac{1}{\gamma\,\sigma\sqrt{L}}, \qquad \mathbb{E}[\mathrm{Loss}] \approx \tfrac{1}{2}\gamma\sigma^2\,\mathrm{Var}(\hat{\pi}) \approx \frac{1}{2\gamma L} \quad \text{(independent of } \mu \text{ and } \sigma\text{!)}$$

Example: $\gamma = 3$, $L = 10$ years gives an expected loss of about $1/60 \approx 1.7\%$ per year.

3 Fixed settings (everyone uses these)

- **Language:** Python 3.10+, with `numpy`, `pandas`, `scipy`, `matplotlib`, `yfinance`.
- **Assets** (Yahoo Finance tickers): `^NSEI` (NIFTY 50, the main asset), `HDFCBANK.NS` (bank), `INFY.NS` (IT), `HINDUNILVR.NS` (FMCG).

- **Period:** 1 Jan 2008 to the latest available date. Daily data, $\Delta t = 1/252$.
- **Risk aversion:** main $\gamma = 3$; robustness checks with $\gamma \in \{2, 5\}$.
- **Risk-free rate:** 91-day Indian T-bill yield (Person 2 provides it). Until then use the placeholder `R_PLACEHOLDER = 0.065`.
- **Random seed:** 42 everywhere, so results are reproducible.
- **Outputs:** figures as PNG (150 dpi) in `figures/`, tables as CSV in `results/`, file names starting with your person number (e.g. `p4_pihat_hist_L10.png`).

4 Repository structure and shared code

```
merton-project/
  README.md
  requirements.txt
  data/raw/                  # downloaded CSVs (Person 2)
  src/
    config.py                # constants above (Person 1)
    theory.py                # formulas from Section 2 (Person 1)
    data.py                  # downloading/loading data (Person 2)
    estimation.py            # mu, sigma, pi estimates (Person 2)
    simulate.py              # Euler-Maruyama simulators (Person 3)
  experiments/
    p2_data_summary.py
    p3_simulator_checks.py
    p4_estimation_risk.py
    p5_backtest.py
  figures/   results/   report/
```

These functions are shared. **Names, arguments and return types must be exactly as below**, because other people's code calls them. If you need to change one, tell Person 1 first.

```
# src/theory.py  (Person 1)
def merton_pi(mu, sigma, r, gamma) -> float          # (mu - r) / (gamma * sigma**2)
def ce_rate(pi, mu, sigma, r, gamma) -> float        # equation (CE) in Section 2.4
def loss(pi_hat, pi_star, sigma, gamma) -> float     # 0.5*gamma*sigma**2*(pi_hat-pi_star
    )**2
def utility(x, gamma) -> np.ndarray                  # x**(1-gamma)/(1-gamma), log(x) if
    gamma==1

# src/data.py  (Person 2)
def load_prices(ticker, start="2008-01-01", end=None) -> pd.Series   # daily adjusted
    close, DatetimeIndex
def load_risk_free() -> pd.Series      # monthly 91-day T-bill rate, annual, continuous,
    decimal
def risk_free_constant() -> float      # average of load_risk_free() over the sample

# src/estimation.py  (Person 2)
def log_returns(prices) -> np.ndarray                    # ln(S_i / S_{i-1}), NaNs
    dropped
def estimate_params(log_rets, dt=1/252) -> tuple         # (mu_hat, sigma_hat)
def estimate_pi(log_rets, r, gamma, dt=1/252) -> float   # merton_pi(*estimate_params
    (...), r, gamma)

# src/simulate.py  (Person 3)
def simulate_gbm(S0, mu, sigma, T, n_steps, n_paths, method="exact", Z=None, seed=None)
    -> np.ndarray
```

```
    # returns array of shape (n_paths, n_steps + 1); method in {"exact", "euler", "
    milstein"}
    # Z: optional pre-drawn standard normals of shape (n_paths, n_steps), to reuse
    randomness
def simulate_wealth(x0, pi, mu, sigma, r, T, n_steps, n_paths, method="euler", Z=None,
    seed=None) -> np.ndarray
    # constant fraction pi; returns array of shape (n_paths, n_steps + 1)
```

Rules for all code (and for any AI writing it): comment every function with what it computes and the formula it uses; no hard-coded numbers except in `config.py`; vectorise with numpy (no Python loops over paths); every experiment script must run from start to finish with one command and regenerate all its figures and tables.

5 Person 1 — Theory and coordination (Rishi)

Goal: own the mathematics, keep everyone's pieces consistent, and assemble the proposal and report.

Tasks

1. Write a clean 2-page derivation note (shared with the group by 27 Oct) covering Section 2 in full detail: self-financing ⇒ wealth SDE; dynamic programming principle and Itô ⇒ HJB; first-order condition; the ansatz; solving for $f(t)$; a short remark on verification (why the HJB solution is the true optimum).
2. Derive the certainty-equivalent formula (4) (lognormal moment), the loss formula, the "worse than cash" condition, and the approximations $\mathrm{sd}(\hat{\pi}) \approx 1/(\gamma\sigma\sqrt{L})$ and $\mathbb{E}[\text{Loss}] \approx 1/(2\gamma L)$.
3. Implement `src/config.py` and `src/theory.py` (Section 4) with a few unit checks (e.g. `ce_rate` is maximised at `merton_pi`).
4. Write and submit the **31 Oct proposal**: title, problem description, model, methodology, expected outcomes, names and IDs, plus first results (Person 2's parameter table, Person 3's accuracy plot).
5. Assemble the final 3–4 page report, write the Introduction, Model and Conclusion, and make sure all methods and datasets are referenced.
6. Run the viva rehearsal (13–14 Nov): everyone explains the derivation once.

Report section: Model and theory (about 1 page).

6 Person 2 — Data and parameter estimation

Goal: provide clean data and the estimation functions everyone else uses, and show that σ is estimated well but μ is not. **Basic code due 30 Oct**, because Persons 4 and 5 depend on it.

Tasks

1. **Prices.** In `src/data.py`, download daily prices with `yfinance` (`auto_adjust=True`, so stock prices include dividends and splits) for the four tickers in Section 3. Save each as `data/raw/<ticker>.csv` so later runs do not re-download. Drop missing values; check there are no zero or negative prices. Note in the report that NIFTY 50 is a *price* index (dividends excluded), which slightly understates its μ.
2. **Risk-free rate.** Get monthly 91-day T-bill yields for India (RBI Database on Indian Economy, `https://data.rbi.org.in`). Save as `data/raw/tbill_91d.csv`. Convert a quoted annual

yield y (in %) to a continuous decimal rate $r = \ln(1 + y/100)$. Implement `load_risk_free()` and `risk_free_constant()`. If the series cannot be obtained, use a documented constant and say so in the report.

3. **Estimation.** In `src/estimation.py`, implement `log_returns`, `estimate_params` and `estimate_pi` using the formulas in Section 2.5 (remember the $+\hat{\sigma}^2/2$ correction; it comes from Itô's lemma).

4. **Table 1 (parameters).** In `experiments/p2_data_summary.py`, for each asset report: sample years L, number of returns n, $\hat{\mu}$ with standard error $\hat{\sigma}/\sqrt{L}$ and 95% CI, $\hat{\sigma}$ with standard error $\hat{\sigma}/\sqrt{2n}$, and $\hat{\pi}$ for $\gamma = 2, 3, 5$. Save to `results/p2_params.csv`.

5. **Rolling estimates.** Using a 5-year window moved forward one month at a time, plot $\hat{\mu}_t$, $\hat{\sigma}_t$ and $\hat{\pi}_t$ over time for each asset (`figures/p2_rolling_<ticker>.png`). Comment: $\hat{\sigma}_t$ is fairly stable while $\hat{\mu}_t$ and $\hat{\pi}_t$ swing wildly.

6. **Is GBM realistic?** For NIFTY, plot a histogram of daily log returns with the fitted normal density, and a normal QQ plot. Report skewness, excess kurtosis and the Jarque–Bera test (`scipy.stats.jarque_bera`). Fat tails mean GBM is only an approximation, which is why Person 4 also uses a bootstrap.

Report section: Data and estimation (about half a page, Table 1 + one rolling plot).
Viva questions to prepare: Why add $\hat{\sigma}^2/2$ to get $\hat{\mu}$? Why does more frequent data help estimate σ but not μ?

7 Person 3 — Simulator and verification

Goal: build the simulation engine everyone uses, show it is accurate, and confirm numerically that π^* maximises expected utility when μ, σ are known. `simulate.py` **due 30 Oct**, because Person 4 depends on it.

The simulator (`src/simulate.py`). Split $[0, T]$ into steps of size Δt; at each step draw $Z \sim N(0, 1)$ independently for each path; then $\Delta W = \sqrt{\Delta t}\, Z$.

- Price, exact: $S_{k+1} = S_k \exp\big((\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma \Delta W_k\big)$.
- Price, Euler–Maruyama: $S_{k+1} = S_k\big(1 + \mu\Delta t + \sigma\Delta W_k\big)$.
- Price, Milstein: $S_{k+1} = S_k\big(1 + \mu\Delta t + \sigma\Delta W_k + \frac{1}{2}\sigma^2(\Delta W_k^2 - \Delta t)\big)$.
- Wealth (constant π), Euler: $X_{k+1} = X_k\big(1 + [r + \pi(\mu - r)]\Delta t + \pi\sigma\Delta W_k\big)$; exact: $X_{k+1} = X_k \exp\big((r + \pi(\mu - r) - \frac{1}{2}\pi^2\sigma^2)\Delta t + \pi\sigma\Delta W_k\big)$.

Test A — Is the simulator accurate? ($\mu = 0.10$, $\sigma = 0.20$, $S_0 = 100$, $T = 1$, 10,000 paths.)

1. Draw Brownian increments once on the finest grid ($\Delta t = 1/1008$). For coarser grids ($1/12, 1/52, 1/252$) *sum blocks of the fine increments*, so every scheme uses the *same* Brownian path as the exact solution. (This is essential; see Higham 2001.)
2. Strong error $= \mathbb{E}\,|S_T^{\text{scheme}} - S_T^{\text{exact}}|$; weak error $= |\mathbb{E}[S_T^{\text{scheme}}] - S_0 e^{\mu T}|$.
3. Plot both against Δt on log–log axes (`figures/p3_convergence.png`). Expected slopes: Euler strong ≈ 0.5, Milstein strong ≈ 1, weak ≈ 1. (Weak error may be hidden by Monte Carlo noise of size about $\sigma S_0/\sqrt{N}$; say so if it happens.)

Test B — Is π^* really the best? (Use NIFTY's $\hat{\mu}, \hat{\sigma}, r$ from Person 2, or placeholders until ready; $\gamma = 3$, $x_0 = 1$, $T = 5$, $\Delta t = 1/52$, 20,000 paths.)

1. For each π on the grid $-0.5, -0.4, \ldots, 2.5$, simulate wealth paths with `simulate_wealth` (Euler). Use the *same* random numbers `Z` for every π (common random numbers) so the

comparison is fair.

2. For each π, compute the average utility $\bar{U} = \frac{1}{N}\sum U(X_T)$ and convert it to a simulated certainty equivalent: $\mathrm{CE}_{\mathrm{sim}}(\pi) = \frac{1}{T}\ln\big(((1-\gamma)\bar{U})^{1/(1-\gamma)}/x_0\big)$.
3. Plot $\mathrm{CE}_{\mathrm{sim}}$ against π together with the theoretical curve `ce_rate`, and mark π^* (`figures/p3_ce_vs_pi.png`). The curve should peak at π^* and match the theory. Repeat for $\gamma = 2, 5$.
4. Plot histograms of X_T for $\pi = 0.5\pi^*$, π^*, $2\pi^*$ to show the risk–reward trade-off.

Test C — Sensitivity of π^*. Using `merton_pi`, plot π^* against $\gamma \in [1, 10]$, $\sigma \in [0.1, 0.4]$ and $\mu - r \in [-0.05, 0.15]$ (`figures/p3_sensitivity.png`). Explain each shape in one sentence (e.g. $\pi^* \propto 1/\sigma^2$).

Report section: Simulation and verification (about half a page, convergence plot + CE-vs-π plot).
Viva questions to prepare: Where does $\sqrt{\Delta t}\,Z$ come from? Why does the Euler error shrink like $\sqrt{\Delta t}$ and Milstein like Δt? Why use the same Brownian path / common random numbers?

8 Person 4 — Estimation risk

Goal: show how unreliable $\hat{\pi}$ is and measure what that costs, in a pretend market where we know the truth.

Idea in one line: we *choose* the true μ_0, σ_0, so the true π_0^* is known (the "answer key"); we generate many fake histories from that market, play the investor who estimates $\hat{\pi}$ from each, and grade every $\hat{\pi}$ against π_0^*.

Experiment A — Parametric (GBM) world

1. **Truth.** Set (μ_0, σ_0, r_0) = full-sample NIFTY estimates from Person 2 (`results/p2_params.csv`); π_0^* =`merton_pi`$(\mu_0, \sigma_0, r_0, \gamma)$ with $\gamma = 3$.
2. **Fake histories.** For each data length $L \in \{5, 10, 20\}$ years: generate $M = 1000$ daily price paths with `simulate_gbm(method="exact", n_steps=252*L)`.
3. **Re-estimate.** For each path: `log_returns` → `estimate_params` → $\hat{\mu}, \hat{\sigma}$ → `estimate_pi` → $\hat{\pi}$; then `loss`$(\hat{\pi}, \pi_0^*, \sigma_0, \gamma)$.
4. **Table** (`results/p4_estimation_risk.csv`), one row per L: mean and sd of $\hat{\mu}$, $\hat{\sigma}$, $\hat{\pi}$; % of $\hat{\pi} < 0$ (shorting); % of $\hat{\pi} > 1$ (borrowing); mean loss (in % per year); % of histories that are *worse than cash* ($\hat{\pi} < 0$ or $\hat{\pi} > 2\pi_0^*$). Also show the benefit $\mathrm{CE}(\pi_0^*) - r_0$ for comparison.
5. **Check the theory.** Compare the observed $\mathrm{sd}(\hat{\pi})$ with $1/(\gamma\sigma_0\sqrt{L})$ and the observed mean loss with $1/(2\gamma L)$ (Section 2.5).
6. **Figures:** histogram of $\hat{\pi}$ for each L with a vertical line at π_0^* (`p4_pihat_hist_L<L>.png`); mean loss against L with the theoretical curve $1/(2\gamma L)$ and a horizontal line at the benefit (`p4_loss_vs_L.png`). Where the loss line is above the benefit line, an investor using estimated parameters is, on average, *worse off than holding cash.*
7. **Simulation check of the loss formula.** For $L = 10$, take the 5th, 50th and 95th percentile $\hat{\pi}$. In the true market, run `simulate_wealth` for $\hat{\pi}$ and for π_0^* ($T = 10$, 20,000 paths, same `Z`), compute $\mathrm{CE}_{\mathrm{sim}}$ as in Person 3's Test B, and confirm the gap matches $\frac{1}{2}\gamma\sigma_0^2(\hat{\pi} - \pi_0^*)^2$.

Experiment B — Bootstrap world (no GBM assumption). Instead of simulating GBM, build each fake history by sampling the *real* NIFTY daily log returns with replacement (length $252L$). The "truth" is the full-sample $\hat{\pi}$. Repeat the table and histograms. Also try a block bootstrap (blocks of 21 days) to keep volatility clustering. If results are similar to Experiment A, the conclusion does not depend on the GBM assumption.

Experiment C — Risk aversion. Repeat the main case ($L = 10$) for $\gamma = 2, 5$. Theory predicts the expected loss scales like $1/\gamma$.

Report section: Estimation risk (about 3/4 page, the table + loss-vs-L plot + one histogram).
Viva questions to prepare: What does "knowing the truth" mean here? Why are the $\hat{\pi}$ so spread out? Why is the expected loss roughly $1/(2\gamma L)$, independent of μ and σ?

9 Person 5 — Real-data backtest

Goal: test whether the Merton strategy with estimated parameters beats simple rules on real Indian market data, using *only information available at the time.*

Procedure (`experiments/p5_backtest.py`)

1. From daily prices (Person 2), build month-end prices and monthly simple returns $R^S_m = S_m/S_{m-1} - 1$. Monthly risk-free return $r^f_m = \exp(r_m/12) - 1$ using `load_risk_free()`.
2. Start once $W = 5$ years of past data are available. At each month-end m:
   (a) take the daily log returns in the past W years *only*, and compute $\hat{\pi}_m =$`estimate_pi`$(\cdot, r_m, \gamma)$;
   (b) hold that fraction for the next month: $R^P_{m+1} = \hat{\pi}_m R^S_{m+1} + (1 - \hat{\pi}_m) r^f_{m+1}$;
   (c) update wealth $X_{m+1} = X_m(1 + R^P_{m+1})$, starting from $X = 1$.

   Assume borrowing and lending at r (state this assumption).
3. **Strategies** (all over the same out-of-sample months): cash ($\pi = 0$); 60/40 ($\pi = 0.6$, rebalanced monthly); all-stock ($\pi = 1$); Merton unconstrained ($\hat{\pi}_m$); Merton constrained ($\hat{\pi}_m$ clipped to $[0, 1]$: no shorting, no borrowing); and, *for reference only*, Merton "oracle" using the full-sample $\hat{\pi}$ (this uses future data, so it is not a real strategy; label it clearly).
4. **Metrics** (annualised from monthly returns): mean return; volatility; Sharpe ratio $\sqrt{12} \cdot \text{mean}(R^P - r^f)/\text{sd}(R^P - r^f)$; maximum drawdown; final wealth; realised certainty equivalent $\text{CE} = 12\,\overline{R^P} - \frac{\gamma}{2} \cdot 12\,\text{Var}(R^P)$ (the empirical version of (4), as in DeMiguel et al. 2009); average $|\hat{\pi}_m|$.
5. **Robustness:** repeat for all four assets, windows $W \in \{3, 5, 10\}$ years and $\gamma \in \{2, 3, 5\}$. Save `results/p5_backtest_summary.csv`.
6. **Figures:** wealth curves (log scale) of all strategies for NIFTY (`p5_wealth_NSEI.png`); $\hat{\pi}_m$ over time with 0 and 1 marked (`p5_pihat_path.png`); bar chart of realised CE by strategy and asset (`p5_ce_bars.png`).

Look-ahead bias: at month m you may only use data up to month m. Using any later data (except in the labelled oracle) makes the results invalid. Mark the 2008 and 2020 crashes on the wealth plot and comment on how each strategy behaved.

Report section: Real-data results (about 3/4 page, metrics table + wealth plot).
Viva questions to prepare: Does estimated Merton beat 60/40? Why might clipping $\hat{\pi}$ to $[0, 1]$ help (link to Person 4's finding)? Why is the oracle not a fair comparison?

10 Timeline

Dates	What must be done
25–27 Oct	Everyone reads Sections 1–4 and their own section. Person 1 shares the derivation note. Person 2 downloads data. Person 3 writes `simulate.py`.
28–30 Oct	Person 2: `data.py`, `estimation.py`, Table 1. Person 3: Test A. Persons 4 and 5: write experiment scripts using placeholder parameters. Person 1: drafts the proposal.
31 Oct	**Proposal submitted** (with Table 1 and the convergence plot as progress).
1–8 Nov	All experiments run on real parameters. Mid-point check-in on 4 Nov.
9–11 Nov	Results frozen. Each person sends their report section (with figures) to Person 1.
12 Nov	Person 1 merges the report; everyone reviews it and checks the code runs end to end.
13–14 Nov	Viva rehearsal: everyone explains the derivation of π^* and their own results.
15 Nov	**Final submission**: code + 3–4 page report.

11 Report outline (3–4 pages)

1. Introduction and research question (Person 1)
2. Model, HJB solution and the loss formula (Person 1)
3. Data and estimation: Table 1 (Person 2)
4. Simulation and verification (Person 3)
5. Estimation risk (Person 4)
6. Real-data backtest (Person 5)
7. Conclusion and limitations (Person 1): constant parameters, fat tails, no transaction costs, borrowing at r, NIFTY as a price index.
8. References

Keep to about 5–6 figures in total; extra figures go in the code repository.

12 Viva quick-check (everyone should be able to answer)

1. **What is the problem?** Choose the fraction π in the stock to maximise $\mathbb{E}[U(X_T)]$.
2. **Why CRRA utility?** The optimal *fraction* does not depend on the wealth level, which gives a clean constant answer.
3. **What does γ mean?** Relative risk aversion, $-xU''(x)/U'(x) = \gamma$.
4. **Where does the HJB come from?** The dynamic programming principle plus Itô's lemma applied to the value function $V(t, X_t)$.
5. **Why is π^* constant?** Constant μ, σ, r and CRRA utility; the ansatz $V = f(t)x^{1-\gamma}/(1-\gamma)$ makes $xV_{xx}/V_x = -\gamma$.
6. **Why is μ hard to estimate?** Its standard error is $\sigma/\sqrt{L}$: only more *years* help, not more frequent data.
7. **What is the certainty equivalent?** The safe yearly return worth the same to the investor as the risky strategy.
8. **When is estimated Merton worse than cash?** When $\hat{\pi} < 0$ or $\hat{\pi} > 2\pi^*$.

9. **What is Euler–Maruyama, and its accuracy?** The SDE version of Euler's method; strong order 1/2 (Milstein: 1).
10. **Main limitations?** Constant parameters, normal returns (real returns have fat tails), no transaction costs, borrowing at the risk-free rate.

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